



LIBRA REACTION WHEELS



PERFORMANCE	LIBRA-065 [NRWA-T065]	LIBRA-2 [NRWA-T2]	LIBRA-6 [NRWA-T6]	LIBRA-80
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FUNCTIONAL CHARACTERISTICS

Torque	20 mNm (Nominal) 24 mNm (Peak)	84 mNm (Nominal) 90 mNm (Peak)	150 mNm (Nominal) 310 mNm (Peak)	175 mNm (Nominal) 300 mNm (Peak)
Momentum	0.65 Nms @ 6500 RPM (Nominal) 0.94 Nms @ 9000 RPM (Peak)	1 Nms @ 1850 RPM (Nominal) 1.47 Nms @ 2700 RPM (Peak)	6 Nms @ 3845 RPM (Nominal) 7.83 Nms @ 5000 RPM (Peak)	80 Nms @ 3600 RPM (Nominal) 93.4 Nms @ 4200 RPM (Peak)
Speed range	±9000 RPM (@ 24 V – 34 V)	±2600 RPM (Speed/Torque mode) ±2700 RPM (Current mode) (@ 24 V – 34 V)	±5000 RPM (@ 20 V – 36 V)	±3000 RPM (@ 20 V) ±4200 RPM (@ 28 V) ±5000 RPM (@ 34 V)
Rotor moment of inertia	$1.0 \times 10^{-3} \text{ kg}\cdot\text{m}^2$	$5.2 \times 10^{-3} \text{ kg}\cdot\text{m}^2$	$14.9 \times 10^{-3} \text{ kg}\cdot\text{m}^2$	$212.5 \times 10^{-3} \text{ kg}\cdot\text{m}^2$
Speed control accuracy	±1 RPM @ >100 RPM	±1 RPM @ >100 RPM	±1 RPM @ >10 RPM ±0.14 RPM @ > 100 RPM	±1 RPM @ >10 RPM ±0.14 RPM @ > 100 RPM

PHYSICAL CHARACTERISTICS

Dimensions (wheel)	102 mm × 102 mm × 105 mm	150 mm × 150 mm × 75 mm	206 mm × 206 mm × 100 mm	460 mm × 190.5 mm
Dimensions (electronics)	Internal	Internal	Internal	198 mm × 182 mm × 36.5 mm
Mass	1.6 kg	2.8 kg	4.7 kg	8.2 kg (wheel) 1.2 kg (electronics)

ENVIRONMENTAL CHARACTERISTICS

Thermal (acceptance)	-10 °C to +45 °C	-10 °C to +55 °C	-30 °C to +55 °C	-30 °C to +55 °C
Mechanical Tests (Qualification)	16.95 g_{RMS} (random)	16.02 g_{RMS} (random)	14 g_{RMS} (random)	Wheel: 14 g_{RMS} (random) Electronics: 24 g_{RMS} (random)
Radiation (TID) (Qualification)	10 kRad (component level)	10 kRad (component level)	30 kRad (component level)	25 kRad (component level)

INTERFACES

Power supply	28 V unregulated	28 V unregulated	28 V unregulated	28 V unregulated
Power consumption	2.6 W (Quiescent) 4 W (Steady state at 3000 RPM) 6 W (Steady state at 6500 RPM)	1.7 W (Quiescent) 4 W (Steady state at 1500 RPM) 4.8 W (Steady state at 2600 RPM)	5.1 W (Quiescent) 15.3 W (Steady state at 3 Nms) 29.3 W (Steady state at 6 Nms) 139.6 W (Peak power consumption)	4.7 W (Quiescent) 17.3 W (steady state at 44 Nms) 28 W (steady state at 86 Nms) 76 W (Peak torque at 32 Nms) 188 W (Peak power consumption)
Data	RS-422	RS-422	Redundant RS-485 (Full Duplex). 460800 baud/s. Up to 4 units sharing data bus.	Redundant RS-485 (Full Duplex). 460800 baud/s. Up to 16 units sharing data bus.
Connector	9-pin D-type Male	9-pin D-type Male	9-pin D-type Male (Power) 25-pin D-type Male (Data)	9-pin D-type Male (Power) 25-pin D-type Male (Data) 25-pin D-type Female (Redundant Data) 15-pin D-type Female (Electronics motor interface) 15-pin D-type Male (Spindle motor interface)

ACCEPTANCE TESTING: All FM parts undergo random vibration (10 g_{RMS}) as well as thermal cycling (four cycles at ambient pressure) to five degrees beyond operational thermal specifications. However, NewSpace can perform additional environmental testing if required by a client.

CONFIGURATION MANAGEMENT: Specifications are subject to change. Please refer to the latest version.

FEATURES

- Wheel torque and momentum capacity suited to client mission requirements
- Torque control available only on Libra-6 and Libra-80.
- Current, Speed and Torque control
- Simple digital interface to the spacecraft bus
- Integration with an optional gyroscope for inertial rate control and inertial angle control modes is possible on T065
- Hermetically sealed (T065 & T2)
- Long demonstrated lifetimes

APPLICATIONS

- High-performance three-axis torque and momentum exchange actuators for agile small satellite missions
- Momentum bias
- Inertial rate control for accurate pointing of imagers

QUALIFICATION

The NewSpace Systems (NSS) Reaction Wheel product range is extensively qualified and based on a robust design with significant in-orbit heritage.

The T065 wheel, which first flew in 2014, has more than 3 million failure-free hours in-orbit with no reported SEUs. Since then, NSS has expanded its wheel range and now offers custom wheel solutions.

With an order book of > 300 wheels, and wheels baselined on 4 constellation programmes, the NSS wheels have become a preferred choice for longer lifetime missions and constellation programmes.



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